

Scalability & Future Plan

Date	03-07-2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Flat-File JSON Storage Layer: Using history.json and feedback.json causes file-locking issues if multiple requests try to write at the same time.	Prevents simultaneous multi-user access and slows down data retrieval as logs grow larger.	High
2	Volatile Local Memory State: App state elements and cache configurations rely on the local host machine's RAM footprint.	Application data contexts and active states reset completely if the terminal session or loopback pipeline closes.	Medium
3	Single-Threaded File Lookups: Searching through logs requires parsing the entire JSON array row-by-row since there is no semantic indexing.	Search and inference speed drops exponentially as thousands of history log frames stack up over time.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
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1	User Load	Handles single concurrent testing instances across loopback ports 8000 and 8501.	Containerize the system using Docker and scale API traffic via an NGINX Reverse Proxy or Kubernetes clusters.
2	Data Storage	Local static flat-files (history.json, feedback.json) bound to local machine directory paths.	Migrate storage layers to a production Vector Database like ChromaDB or Pinecone for handling massive text data matrices.
3	Performance	Basic procedural string search operations executed across plain text variables.	Integrate Multi-Agent RAG Frameworks (such as Agno or LangChain) paired with asynchronous task workers like Celery to run complex AI jobs.
4	Security	Open HTTP server communication with zero validation keys or client authorization guards.	Enforce standard OAuth2 with JWT (JSON Web Tokens) endpoint authorization layers and wrap data paths using HTTPS.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Vector DB Migration & Semantic Search	Month 1 – 2	Replaces slow file parsing with high-speed semantic matching across historic user interactions.
Phase 3	Multi-Agent Framework Integration	Month 3 – 4	Orchestrates multiple dedicated AI agents to execute deep web-scraping fact checks and customized icebreaker workflows simultaneously.
Phase 4	Cloud Microservices & Public Live URL	Month 5 – 6	Deploys containerized modules across cloud-native environments (like AWS or Vercel), offering robust, public access to thousands of live users.